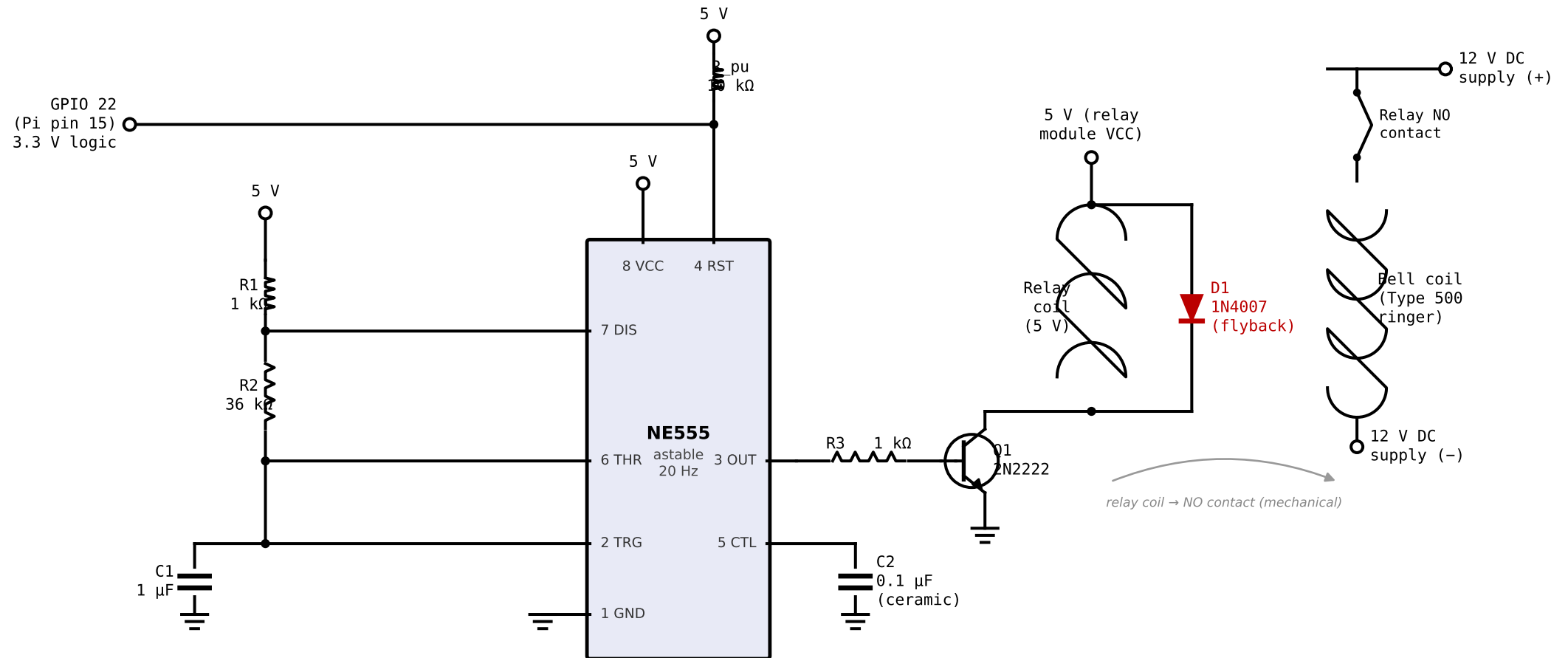


Bell Wiring — Option 2: 555 Astable at 20 Hz, Gated by GPIO 22

Authentic mechanical ring — clapper oscillates at 20 Hz during ring cadence



555 astable frequency: $f = 1.44 \div ((R1 + 2 \times R2) \times C1) = 1.44 \div (73\,000 \times 0.000001) \approx 19.7\text{ Hz}$ (nominal 20 Hz)

GPIO 22 HIGH → RESET (pin 4) released → 555 oscillates → 20 Hz square wave on OUT → Q1 switches relay at 20 Hz → clapper oscillates → authentic bell ring.

GPIO 22 LOW → RESET held LOW → 555 output frozen LOW → relay open → bell silent. Ring cadence (2 s on / 4 s off) is still controlled by the script.

GPIO 22 is 3.3 V logic; 555 RESET LOW threshold $\approx 0.7\text{ V}$ for 5 V supply – direct connection works, no level-shifter needed.

R_{pu} (10 kΩ): pull-up from 5 V to RESET – keeps RESET HIGH if GPIO floats at boot, preventing spurious ringing during startup.

Q1 (2N2222 NPN): R3 = 1 kΩ base resistor. Handles relay coil current $\leq 100\text{ mA}$. Substitute 2N3904 for coils $\leq 50\text{ mA}$.

D1 (red) – flyback diode: ESSENTIAL across relay coil; cathode (banded end) toward 5 V rail. Without it Q1 fails instantly on de-energise.

C2 (0.1 μF ceramic): bypass cap on 555 pin 5 (CTL) – suppresses VCC noise coupling into timing; place as close to IC as possible.

Tune frequency: R2 = 27 kΩ → $\approx 25\text{ Hz}$; R2 = 47 kΩ → $\approx 15\text{ Hz}$. Bell electromagnet is rated 20 Hz; stay within 16–25 Hz for reliable ringing.